WASP-11b

R-0777

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-10_230104` · created 2026-07-28T05:46:51+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459948.8283 +/- 0.0031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.107 +/- 0.05
Transit depth (Rp/Rs)^2	1.14 +/- 1.06 [%]
Orbital Inclination (inc)	84.77 +/- 1.7
Airmass coefficient 1 (a1)	0.87 +/- 0.23
Airmass coefficient 2 (a2)	0.066 +/- 0.036
Scatter in the residuals of the lightcurve fit is	26.27 %
Best Comparison Star	None
Optimal Aperture	0.75
Optimal Annulus	3.0
Transit Duration (day)	0.033 +/- 0.036

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.107 ± 0.05 vs 0.131 (deviation 0.47σ)
Duration fit vs published	0.033 d vs 0.1075 d (deviation 2.02σ)
Mid-time O–C	7.1 min
Depth SNR	2.1
Residual scatter	26.27 %

Light curve

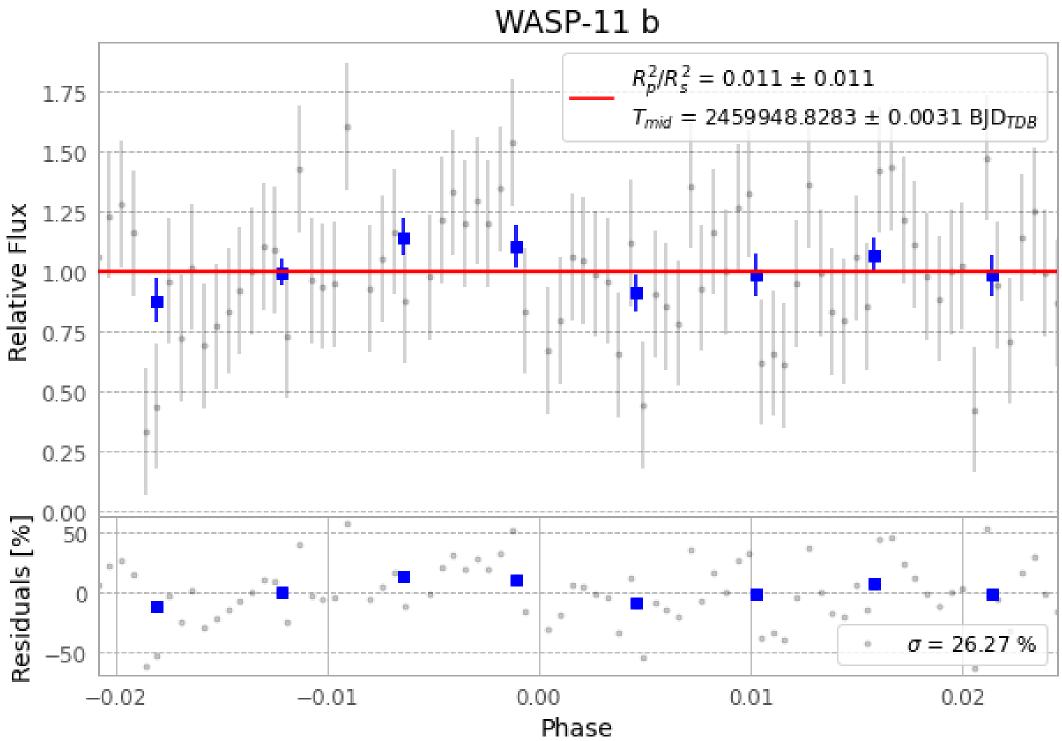

Figure 1 — FinalLightCurve_WASP-11 b_2023-01-03.png (SHA-256 4828a3d902c51198…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [449, 323], 2 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 16c9306fcdbda250…).
`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

82 FITS frames (54 MB), `HATP-10230104055415.FITS` → `HATP-10230104095715.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-230104.log (1864a2a9f9fc…), FinalLightCurve_WASP-11 b_2023-01-03.png (4828a3d902c5…), FinalLightCurve_WASP-11 b_2023-01-03.csv (61e2ba80fa70…)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)